The effect of share repurchases on corporate investment – evidence from the Nordics

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1. Introduction

During the last decades, firms have started to repurchase their own stock on the open market as an alternative way to redistribute cash to shareholders compared to normal cash dividends. Firms repurchase shares on the stock exchange using the firm's cash and then proceed to cancel the bought shares, reducing the number of outstanding shares. Even though no cash are paid to shareholders' bank accounts, it creates value by reducing the total number of outstanding shares which then allows the remaining shares to be worth a larger share of the firm and entitled to a larger share of the profits.

Lately, a criticism towards repurchases has emerged as insiders may use these programs to trigger earnings-per-share bonus programs or spend corporate earning in an inefficient way by repurchasing shares instead of taking advantage of available investment opportunities to facilitate the long-term growth of the firm. If firms choose to sacrifice investment in favour of repurchasing to please shareholders, this could not only have a negative long-term effect for the business growth of individual firms, but also have a negative impact on the economy as a whole as many of the listed firms are some of the largest employers. Investing firms tend to need to start hiring new workers in order to expand, which creates a demand for both skilled and unskilled labour. Lazonick (2014) states that firms listed on the S&P 500 spent 54% of their earnings on repurchase programs and another 37% on cash dividends, leaving a small portion left for corporate investment or increased wages for already hired workers.

The phenomenon of firms sacrificing investment in favour of repurchases has been vastly studied using US data and recent research supports the conclusion that firms are sacrificing investment in favour of repurchases (VanDalsem, 2019; Almeida et al., 2016).

This study aims to investigate if a similar relationship exists in the Nordic region, where very little research has been done in this subject. Do Nordic firms behave in a similar way, prioritizing short-term gains for shareholders by executing share repurchases beyond the optimal level or are they repurchasing due to lack of growth opportunity?

Repurchases has yet to become as popular as it has been for US firms the past few years, which creates the expectation that Nordic firms may not be as willing to make the sacrifice for increased repurchases, due to less pressure from shareholders. This means that the effect for repurchases on investment is expected to be smaller than in similar studies in the US. This paper adds to current literature by extending this question to the Nordic region, which is of

relevance to policy makers, as well as investors. For policy makers, this type of research is crucial in order to understand if firm insiders use repurchases for reasons other than redistributing cash to shareholders, in order to form adequate policies. Investors usually want to maximize the growth of the firms they are investing in, which makes this type of research important in order to be aware of potential skewed incentives.

Using data from the main stock exchanges in Sweden, Norway, Denmark and Finland. The sample consists of 800 firms studied from the beginning of 2000 through the end of 2019. The main outcome variables proxying corporate investment are capital expenditures, change in employment and R&D expenses. This thesis will study the relationship between these variables using both value of repurchases and a binary variable reflecting the decision to repurchase. A Heckman (1979) model will be used to deal with the issue of sample selection bias. The findings of this paper are not consistent with previous studies made on US data and show no evidence that the Nordic firms included in this sample does sacrifice corporate investment in favour of repurchases. The evidence suggests that repurchases are made after taking advantage of available investment opportunities. This is consistent with the hypothesis as repurchases has not become as popular in the Nordic countries, the firms do not face the same pressure to repurchase in order to please shareholders.

The rest of this paper is organized as follows. Part two covers previous research regarding share repurchases, as well as research made for relevant variables and theory used in this paper. Part three covers the data selection and methodology used to identify the effect of interest. Part four presents the descriptive statistics and the main results of the study. In part five the specification is tested for robustness and part 6 offer the concluding remarks.

2. Previous Literature

Almeida et al. (2016) employs a regression discontinuity design to find effects of repurchases on firm investment outcomes. The discontinuity exploited is that firms that are on track to just miss their earnings-per-share targets are much more likely to spend corporate money on repurchases compared to firms that are to just beat their target. Manager compensation programs are often conditioned on meeting specific targets, such as earnings-per-share. The corporate insiders use repurchases programs as a way to trigger these compensation programs by increasing the size of repurchases if the firm is close to meeting targets. The authors conclude

that managers are willing to trade increased investments and increased employment for repurchases that allow them to meet analyst earnings-per-share forecasts.

VanDalsem (2019) recently studied if family owned businesses are different from other firms when it comes to reducing corporate investment in favour of repurchasing more shares and employs a two stage Heckman (1979) approach to purge the sample from selection bias. Family firms are defined as when a member of the founding family has a prominent role in the firm, either CEO or member of board, or has a significant equity stake. The study concludes that family owned firms are less likely to repurchase shares and when they do, it's less likely to be associated with decreases in corporate investment compared to non-family owned firms. In contrast to these findings, Bhagwat and Bruine (2018) find no support for the thesis that firms repurchase on the expense of capital expenditures and argues that firm prioritize investment and then proceed to pay out whatever is left of the profit to shareholders.

According to the agency theory firms that do not face attractive investment opportunities usually spends their spare cash investing in projects that are negatively yielding and reduce the value of the firm. The free cash flow hypothesis was developed by Jensen (1986) where he argues that repurchases is an instrument to manage cash available to managers when firms go into a more mature phase with shrinking investment opportunities. The theory predicts that repurchases should be associated with less profitability, less investments and less cost of capital. Chu & Liu (2016) studies investment in the real estate market and finds that firms that have high cash flows or higher cash reserves tend to pay more for new investments in real estate.

Grullon & Mikealy (2004) studies the information content signalled by the initiation of share repurchase programs and finds that they are not an indication of increased operating performance but associated by a significant reduction in systematic risk and cost of capital. They also conclude that repurchase programs are much more likely to be initiated by firms that lack attractive investment opportunities. Their findings are also consistent with the free cash flow hypothesis as they find that repurchasing firms spend less on investment and reduce their cash reserves by increasing pay-outs.

Firms that have excess cash after making attractive corporate investments, should redistribute these to shareholders rather than making negatively yielding investments. Previous literature has faced difficulties observing "Investment Opportunities" as it is typically known only by

insiders. To tackle this problem several other studies (VanDalsem, 2019;Almeida et al., 2016;Boudry et al., 2013) has used Tobin' Q to measure investment opportunities. Tobin's Q is the ratio of a firm's market value and replacement cost of assets and hence reflects the average realized return on the firm's capital. Lang and Litzenberger (1989) show that this can be used to find over- or underinvesting firms. Tobin's Q will be less than one for firms if the current replacement value for all assets is greater than the firms market value. If a firm's investments are increasing with size and are decreasing in marginal efficiency, Tobin's Q will be less than one for over-investing firms. Firms that still have opportunities to invest with positive yield, will have a ratio of higher than one. Some studies have tried to create advanced algorithms to calculate Tobin's Q (Lindenberg and Ross, 1981; Perfect and Wiles, 1994; Lewellen and Badrinath, 1997). However, Erickson and Whited (2006) found when evaluating these algorithms that they tend to lower the number of observations and introduce other issues such as sample selection bias. They later conclude "Researchers are just as well off using a simple measure of q as using a computationally complex measure".

De Cesari et al. (2012) investigates timing of repurchase transactions based on ownership structure. This study also identifies heterogenous effects among investor types where institutional owners are less successful to purchase own shares at a bargain price compared to insider owners. Palladino (2020) show some evidence that insiders tend to time their personal sale of shares with repurchase programs. She concludes as managers do not have to announce dates of repurchase, they can easily time them for own personal gain. Additionally, Kahle (2002) finds that repurchases increase with the use of options in employee and executive compensation plans.

A common reason stated by firms to start repurchasing shares are to avoid dilution effects, as firms commonly use issuances of new shares to compensate managers and staff, as well as to raise capital, shareholders are often diluted over time. Root, Rozycki & Suh (2014) studied share repurchases and long-term dilution and found that despite repurchases, 71% of their sample was diluting their outstanding shares instead of increasing share value over time.

To repurchase shares has also been shown to have some advantages to the corporation in comparison to increasing their cash dividends. Brav et al. (2005) interviewed 384 CFOs and Treasurers and found that management are much more reductant to cut dividends as investors find them more tied to long-term earnings. Rather than increasing dividends managers prefer

to increase pay-outs during high profit periods by repurchasing shares, as repurchase programs are more flexible and permits better opportunity to optimize current investment needs. This is also supported by Floyd et al.'s (2015) which study showed that after the financial crisis in 2000, the dollar value of repurchases fell by 35% compared to cash dividends which fell by 6.5%.

Previous literature has identified numerous reasons other than "just" returning capital to shareholders as a motive to repurchase shares. The studies that inspired this thesis also find significant results for repurchases on investment variables while controlling for investment opportunity. This sparks the question that this may be the case outside of the US as well which will be the scope of this thesis. According to the findings of Erickson and Whited (2006), a simple approach to identifying investment opportunity will be utilized.

3. Methodology

3.1 Data and sample selection

The data consists of 800 firms over 20 years, reaching from the beginning of 2000 to the end of 2019. The included firms are traded on the major stock exchanges in Sweden, Norway, Denmark and Finland and is extracted from the Thompson Reuters database Datastream. As per Almeida et al. (2016) firms with missing value for assets are excluded as this makes construction of necessary variables impossible. The final sample consists of 11767 firm-year observations and as the different exchange's reports data in the respective local currency, all values are converted to a constant SEK value.

3.2 Variables

Following Almeida et al. (2016), three different variables will be analysed in order to identify investment; capital expenditures, employment and R&D expenses. Capital expenditures are the total expenses for new machines, the development of new services or any other action made in order to increase total sales or profits. Employment is measured by the change of the workforce in each firm and research and development (R&D) is the change in spending on developing new products.

Repurchases will be constructed as per Fama and French (2001) who measures net purchases as the increase is the treasury stock for firms where the value is not zero or missing. If the firm has a net issuance of shares for the fiscal year, repurchases are set to 0. The value of repurchases are then divided by the value of total assets for the prior year, the lagged value of assets. A binary variable for repurchases will also be created, *RepurchaseDum*, where the value is set to one if the firm repurchased for the firm-year observation and zero otherwise.

To adequately test the hypothesis, we need to identify which firms face attractive opportunities to invest excess cash flows and which do not, this study will according to previous literature use Tobin's Q as a proxy for "Investment Opportunities". Tobin's Q is defined;

$$Tobin's\ Q_i = \frac{Market\ Value_i + Preferred\ Stock_i + Debt_i}{Total\ Assets_i} \qquad (1)$$

where market value is stock price multiplied by total number of outstanding shares. Preferred Stock represents the liquidation value of all outstanding preferred shares and debt is the book value of debt. Total assets are the book value of assets. According to the findings of Lang and Litzenberger (1989), Tobin's Q can be used to identify over- or underinvesting firms. Tobin's Q is the ratio of a firm's market value and replacement cost of assets and reflects the average realized return on the firm's capital. The Tobin's Q ratio will be less than one for firms if the current replacement value for all assets is greater than the firms market value. Assuming firm's investments are increasing with size and are decreasing in marginal efficiency, Tobin's Q will be less than one for over-investing firms. Firms that still have opportunities to invest with positive yield, will have a ratio of higher than one.

Following Lang et al. (1991), firms that have cash flows higher than the sample median and Tobin's Q below one is identified as firms with free cash flows and hence more likely to repurchase shares. Other firm characteristics that may be influencing a firm's probability to repurchase shares and included as controls are leverage and cash (Dittmar, 2000). Similar to VanDalsem (2016), the variable SDCashFlow measures the standard deviation of cash flows as firms with more predictable cashflows should have a higher probability of repurchasing

compared to firm with less predictable cash flows as they do not need to save as much cash reserves for the unpredictable future.

Cash is the value of cash and cash equivalents plus depreciation normalized by the value of lagged assets. Leverage is the firm's total book value of debt divided by lagged assets. Following VanDalsem (2019), size is used to control for the size effect and is measured by the log of assets. Return on Assets (ROA) is defined as the firm's net income divided by its total assets and the purpose of ROA is to control for how efficient a firm in allocating its assets to generate earnings. EBIT is the Earnings Before Interest and Taxes, also normalized by the lagged value of assets. The variables that are scaled by the value of assets is multiplied by 100 to allow interpretation as percentage points.

The control variables used in the selection equation is HighCashFlow and ROA which capture the decision firm's make whether to repurchase shares or not. HighCashFlow are defined as the firms which have cash flows above the sample median and a Tobin's Q below 1 to account for their opportunities to invest their high cash flows in a profitable way.

3.3 Empirical Approach

Each of the outcomes for investment will initially be fit to a fixed effects model using *RepurchaseDum* as the main explanatory variable, specified as:

$$Investment_{i,t} = Firm_i + Year_t + \beta_1 RepurchaseDum_{i,t} + \beta_2 Tobin's\ Q_{i,t} + \beta_3 Cash_{i,t} + \beta_4 Leverage_{i,t} + \beta_5 Size_{i,t} + u_{it} \quad (2)$$

This specification provides insight if the decision to repurchase has any effect on the investment variables. Theory suggests that the relationship between investment variables and the decision to repurchase should be zero, while controlling for availability of cash and investment opportunity. Managers decision to repurchase should be based on how much cash the firm has available and which investment opportunities they have to spend that cash on. These effects are captured in model 2 by β_2 and β_3, and a significant and negative $\beta_1$would indicate that firms make the repurchase decision based on other variables and are sacrificing investment for repurchases. A significant β_4 would indicate that firms take on debt in order to finance

investment and β_5 is expected to be negative as larger, more mature firms have less opportunity to make profitable investments (Jensen ,1986).

A positive relationship between the decision to repurchase and investment would indicate that high investing firms are also spending cash on repurchases. In order for this scenario to occur, firms need to have access to a large reserve of cash, as well as having some investment opportunity. They would then spend the necessary amount to take advantage of opportunities, and then proceed to repurchase using the leftovers. This effect should however be captured by β_3 or possibly β_4 if firms are willing to finance the repurchases by taking on additional debt.

Given the characteristics of this analysis, variables included are likely to suffer from multicollinearity as for example, a high leverage may lead to a high cash reserve or lower levels of cash may due to a high Tobin's Q. An issue for this analysis would be if repurchases was highly correlated with the other independent variables, as this would lead to inconsistent results. If repurchases are colinear with the other independent variables, it makes interpretation of final results difficult as it's impossible to keep other variables constant when repurchases change – a change in repurchases is always associated with a change in another independent variable. To control for this, this thesis will perform a variance inflator factor test. Testing both the variable measuring the level of repurchases and the binary variable for multicollinearity with the other independent variables, the test shows no signs of multicollinearity that would cause issues for the analysis (appendix 2).

As firms do not flip a coin in order to make the decision whether to repurchase shares or not, a model that attempts to estimate the level of repurchases, would likely be biased if not adjusted for certain firm characteristics which puts the firm either the repurchasing group or not. This means that when trying to estimate the effect using value of repurchases as the main independent variable, the sample will be heavily reduced as only a fraction of the sample is actually repurchasing. According to theory, firms should be repurchasing if they lack attractive investment opportunity, which would directly cause the reduced sample on only consist of low investing firms and introduce a negative bias to the final estimations.

In order to solve the sample section issue, this study will utilize a two stage Heckman (1979) model to adjust for the selection bias. To provide a theoretical explanation of this methodology,

the Heckman model assumes a relationship between *Investment* and explanatory variables according to:

$$Investment_i = \beta X_i + u_{1i} \tag{3}$$

where *Investment* is our main variable of interest and X is the full set of explanatory variables, β is a set of unknown parameters and u_{1i} is the error term. However, the investment outcome will only be estimated for firm i if it does engage in repurchasing, so for firms where:

$$\gamma Z_i + u_{2i} > 0 \tag{4}$$

Z is the same full set of the explanatory variables, γ is the set of unknown parameters and u_{2i} is the error term. The variables used in the selection equation then essentially serves as an instrument, attempting to adjust for the endogeneity problem with firms self-selecting into the final sample by deciding to repurchase. The first stage of the model is to estimate the probability of the individual firm to engage in repurchasing activities, which is a probit model expressed:

$$Prob(R = 1|Z) = \Phi(\mathrm{Z}\gamma) \tag{5}$$

where R = 1 if the firm repurchases, Z is the set of variables which firms base their decision on, γ is the set of parameters and Φ is the cumulative distribution function. The first stage is a probit model where the variables used in the second stage is included, as well as additional variables that could explain the decision to repurchase but explain little of the level. The result of the first stage probit selection equation tris to estimate γ from equation 4, the parameters are then used to calculate the inverse mills ratio which is the probability density function divided by the cumulative density function ratio and denoted by lambda.

$$\lambda = \frac{\phi(Z'\gamma)}{\Phi(\mathrm{Z}'\gamma)} \tag{6}$$

which essentially is the probability that the firm decides to repurchase divided by the cumulative probability of the firm's decision. The inverse mills ratio is then included as an explanatory variable in the second stage of the model which is a standard two-way fixed effects regression containing both firm and year fixed effects.

The formal identification from the model comes the assumption that if we could observe a fully balanced dataset with observations of repurchases for all firms, it could be estimated by a least square's regression. This is possible if the assumption of normality is fulfilled, meaning that if all firms were repurchasing, they would be normally distributed.

The first stage of the Heckman two step in this analysis, consists of a probit model attempting to identify the determinants of firm's decision to repurchase or not. Previous literature identifies free cash flows as the main determining factor whether the decision to repurchase or not is made, denoted *HighCashFlow* in equation 7. These firms have available cash without facing attractive investment opportunities and seen as candidates to repurchase shares. The first stage of the Heckman model is specified:

$$\begin{aligned} RepurchaseDum &= Firm_i + Year_t + \alpha_1 HighCashFlows_{i,t} + \alpha_2 SDCashFlow_{i,t} \\ &+ \alpha_3 Tobin's\ Q_{i,t} + \alpha_4 Cash_{i,t} + \alpha_5 Leverage_{i,t} + \alpha_6 Size_{i,t} + u_{it} \end{aligned} \tag{7}$$

From the first stage model, HighCashFlows is expected to offer the large part of explanatory power as this is what has been seen in previous studies and consistent with theory. Firms that experience small deviations in their cash flow are expected to have higher repurchases as they easier can predict future cash flows and require a smaller cash reserve. The sign for Tobin's Q should be negative as less investment opportunity should be associated with a higher probability to repurchase and Cash is expected to be positively correlated with the decision to repurchase. Leverage controls for the fact that some firm may be willing to take on debt in order to repurchase.

The second stage of the Heckman twostep is the main specification and is including the inverse mills ratio to purge the model from sample selection bias:

$$\begin{aligned} Investment_{i,t} = {} & Firm_i + Year_t + \beta_1 Repurchase_{i,t} + \beta_2 Tobin's\ Q_{i,t} \\ & + \beta_3 Cash_{i,t} + \beta_4 Leverage_{i,t} + \beta_5 Size_{i,t} + \lambda_{i,t} + u_{it} \end{aligned} \tag{8}$$

which is including the level of repurchases as the main variable of interest. This equation will capture if firms trade of investments in favour of repurchases at which level. Tobin's Q will capture the effect if firms spend their cash on investment due to lucrative opportunity, cash will capture if high levels of investment is due to high levels of spare cash. Size control for the firm

size as larger companies are expected to spend less on investment due to being in a more mature phase. Leverage controls for firms that take on additional debt in order to finance investment.

4. Results

4.1 Descriptive Statistics

Table 1 shows total number of observations where a repurchase is occurring by year and the total SEK value of repurchases. The data shows a clear trend that repurchases are becoming more and more common way for firms to return cash to shareholders. There is no clear trend in the value of the repurchases, which is rather showing a decrease in money spent on total repurchases with 2005 being the peak year. Similar US studies (VanDalsem, 2019; Almeida et al., 2016) show that repurchasing is much more common among US firms compared to those included in this sample.

Table 1. Share of firms repurchasing by year.

	Firms Repurchasing	Value of Repurchases (MSEK)
2000	8.6%	112.02
2001	8.6%	54.72
2002	7.6%	61.16
2003	7.2%	120.6
2004	7.1%	148.29
2005	7.6%	230.86
2006	7.2%	150.43
2007	8.2%	191.13
2008	10%	159.69
2009	10.9%	88.02
2010	10.9%	90.57
2011	12.2%	66.12
2012	13.1%	48.32
2013	14.1%	44.51
2014	14.4%	57.44
2015	15.2%	64.12
2016	15.4%	66.68
2017	14.6%	67.67
2018	15.9%	46.54
2019	7.1%	174.66

Shows the percentage of firms making at least one repurchase during any fiscal year dating the observed time period. Including the full sample of 800 firms and 16.000 firm-year observations.

While this study's sample see repurchases in the range of roughly 7-16% each year, VanDalsem (2019) see ranges between 65-82% of firms using repurchases. The drop in firms repurchasing in 2019 is unclear and a drop of repurchases around the financial crisis would be expected due to increased uncertainty.

Table 2 describes all the variables included in the analysis for all firm-year observations and table 3 shows the same variables divided between repurchasing and non-repurchasing firms. Repurchasing firms spend more on capital expenditures and have more employees compared to non-repurchasing firms, which however spend significantly more on R&D. Both types have similar Tobin's Q, indicating on average similar investment opportunities. Repurchasing firms show significantly higher Return on Assets, making them much more efficient at generating profit from their assets. Regarding both Tobin's Q and Return on Assets, the descriptive statistics show some unexpected results as theory would predict the opposite to be true, where non-repurchasers should have higher investment opportunity (Jensen, 1986).

Table 2. Descriptive Statistics

	Mean	Median	St.Dev
Capex/Assets (%)	4.81	2.72	7.25
Employees/Assets mil	.78	.4	3.78
R&D/Assets (%)	7.21	2.27	21.48
Repurchases/Assets	.37	0	1.65
Tobins Q	1.22	.5	3.43
Return on Asset (%)	11.39	6.16	73.05
Assets (MSEK)	96641.37	2642	1400000
Market Cap (MSEK)	22872.88	1914.28	78784.48
EBIT/Assets (%)	1.21	6.07	83.21
Cash Flow/Assets (%)	4.31	6.59	25.3
Leverage (%)	24.87	21.94	21.9

Shows the summary statistics for the full sample of 800 firms and 16000 firm year observations. Variables are scaled by the value of lagged assets.

Repurchasing firms are also much more profitable and have higher cash flows, which is expected. The data shows firms to be either high in investment - high cashflows or in low investment – low cash flows and it is not explained by debt. This could be an effect of repurchasing firms being much larger and in different stages compared to their non-repurchasing counterparts.

Table 3. Differences in means between repurchasing and non-repurchasing firms.

	Non-Repurchasing Firms		Repurchasing Firms	
	Mean	Median	Mean	Median
Capex/Assets (%)	4.72	2.52	5.35	3.69
Employees/Assets mil (%)	.74	.4	.99	.39
R&D/Assets (%)	7.83	2.39	4.6	1.57
Value of Repurchases/Assets	0	0	1.39	.27
Tobins Q	1.23	.5	1.18	.46
Return on Assets (%)	9.52	5.97	22.22	7.9
Assets (MSEK)	107845.8	2295.55	29180.99	5068.68
Market Cap (MSEK)	20564.12	1661.68	36431.16	3746.63
EBIT/Assets (%)	.4	5.97	6	6.64
Cash Flow/Assets (%)	3.67	6.28	8.17	8.01
Leverage (%)	24.67	21.58	26.08	23.91

4.2 Multivariate Results

Table 4 shows the results from the fixed effects model for each of the investment measurements using the dummy for if a firm performed a repurchase or not as the independent variable. The coefficient for repurchased is only significant at the 10% level for the outcome on employment. It does show the expected sign that repurchasing is associated with a lower number of employees compared to non-repurchasing firms. The number should be interpreted as 0.12 less employees per million of assets for repurchasing firms.

The coefficient for the R&D and Capex models does not show any level of significance for the repurchase variable. The model does indicate that firms decide spending on R&D based on other variables.

Table 4: Fixed effects models identifying relationship between the decision to repurchase on investment variables.

	(1)	(2)	(3)
	Capex	Employment	R&D
RepurchaseDum	-0.578	-0.120*	0.347
	(2.344)	(0.068)	(0.535)
Leverage	0.318	0.004**	-0.012
	(0.312)	(0.002)	(0.022)
Size	11.503	-0.239***	-3.122***
	(11.268)	(0.037)	(0.936)
Tobin's Q	2.577	0.022*	0.215***
	(1.807)	(0.012)	(0.082)
EBIT	2.193*	0.001**	-0.140***
	(1.193)	(0.000)	(0.046)
Cash	-0.345	0.000	0.072***
	(0.234)	(0.000)	(0.024)
Constant	-271.732	6.031***	76.140***
	-0.578	-0.120*	0.347
Firm Fixed Effects	YES	YES	YES
Year Fixed Effects	YES	YES	YES
R-squared	0.012	0.046	0.288

Standard errors are in parenthesis

**** p<0.01, ** p<0.05, * p<0.1*

Larger firms are spending less compared to smaller firms on R&D and employment and the coefficient for Tobin's Q has a positive relationship, meaning that firms invest more if they face better opportunities. Table 5 shows the first stage of the two stage Heckman model where the binary variable for repurchases is set as the dependent variable. The purpose of this equation is to estimate the inverse mills ratio in order to adjust for the sample selection bias that would otherwise occur. The difference in the results, despite regressing the same variables on the same binary outcome, is due to Stata using different samples as a result of some missing observations in the different variables.

The most important factor that determines firm's decision to repurchase is high cash flows, where firms above the median are much more likely to initiate repurchase programs. Firms are also more likely to repurchase given that they have a high return on assets.

Table 5: First stage hackman results: Probit regressions for each sample identifying determinants of probability to repurchase.

	(1)	(2)	(3)
	RepurchaseDum (Capex)	RepurchaseDum (Employment)	RepurchaseDum (R&D)
High Cash Flows	0.233***	0.257***	0.291***
	(0.047)	(0.048)	(0.061)
SD Cash Flow	-0.000*	-0.000*	-0.000
	(0.000)	(0.000)	(0.000)
Return on Assets	0.001***	0.001***	0.001***
	(0.000)	(0.000)	(0.000)
Leverage	-0.000	-0.000	-0.006***
	(0.000)	(0.000)	(0.001)
Size	0.041***	0.043***	0.076***
	(0.012)	(0.012)	(0.014)
Tobin's Q	0.007	0.009*	0.026***
	(0.005)	(0.005)	(0.005)
Constant	-2.002***	-2.049***	-3.226***
Year Fixed Effects	YES	YES	YES

Standard errors are in parenthesis

**** p<0.01, ** p<0.05, * p<0.1*

Table 6 is the main results where the sample selection bias from only including observations where a repurchase happened has been corrected for by including the inverse mills ratio. The inverse mills ratio is denoted by Heckman's lambda in the regression table and is significant at the 1% level with a large coefficient for Capex and R&D. For employment the coefficient is small and insignificant indicating that no selection bias would occur without the correction. The adjustment is consistently negative together with large coefficients indicating that the sample is negatively selected, and the unadjusted sample would have been largely biased downwards. The main variable of interest, which is the value of repurchases, does show some unexpected results.

Table 6: Two step Heckman: Fixed Effect models including the inverted mills ratio to control for sample selection.

	(1) Capex	(2) Employment	(3) R&D
Repurchase	0.339	0.014*	0.338***
	(0.211)	(0.007)	(0.066)
Leverage	-0.004	0.003***	0.021
	(0.025)	(0.001)	(0.015)
Size	-5.199***	-0.098**	-0.860**
	(1.231)	(0.041)	(0.399)
EBIT	-0.189***	-0.002	-0.022
	(0.051)	(0.002)	(0.019)
Cash	0.201***	0.005***	0.113***
	(0.029)	(0.001)	(0.007)
Tobin's Q	-0.138	0.008	-0.176*
	(0.324)	(0.011)	(0.091)
Return on Assets	-0.007	-0.000	-0.007*
	(0.013)	(0.000)	(0.004)
Constant	164.724***	2.445**	46.725***
	(35.538)	(1.113)	(11.397)
Heckman's lambda	-22.57***	-0.35	-5.98***
	(8.27)	(0.23)	(2.06)
Firm Fixed Effects	YES	YES	YES
Year Fixed Effects	YES	YES	YES

Standard errors are in parenthesis
**** p<0.01, ** p<0.05, * p<0.1*

The coefficients are consistently positive which is not in line with previous studies made on US data and is a clear indication that firms do not sacrifice corporate investment in favour of satisfying shareholders by repurchasing shares on the open market. The coefficient for capex is not significant at any level, but the model does show some effects on employment and R&D which are significant at the 10% and 1% level, respectively. This shows that repurchases are associated with higher investment levels for the firms that are included in this sample. Both measurements that proxy attractiveness for spending on investment, Tobin's Q and Return on Assets, show a negative relationship with R&D and cash has a positive relationship. The results

indicate that repurchases are made by firms in a more mature phase and are not sacrificing investment for repurchases.

5. Robustness

As repurchases are much more uncommon in this study's sample compared to similar US studies, the estimates have a higher risk of being driven by large outliers. In order to check for this, the same models will be estimated with the first and tenth decile omitted from the sample. In table 8 (appendix) the first stage results are presented.

Table 7: Two step Heckman: Fixed Effect models including the inverted mills ratio to control for sample selection. Decile 1 and 10 is omitted.

	(1) Capex	(2) Employment	(3) R&D
Repurchase	0.48	0.04	-0.77
	(3.67)	(0.13)	(1.41)
Leverage	0.07***	0.00***	-0.00
	(0.02)	(0.00)	(0.01)
Size	-2.58*	-0.16***	-0.87
	(1.56)	(0.05)	(0.66)
EBIT	0.09*	0.00	-0.01
	(0.05)	(0.00)	(0.02)
Cash	0.02	-0.00	0.09***
	(0.05)	(0.00)	(0.02)
Tobin's Q	0.21	-0.04***	-0.44**
	(0.48)	(0.02)	(0.18)
Return on Assets	-0.01	0.00	-0.00
	(0.01)	(0.00)	(0.00)
Constant	92.53*	4.69***	46.27***
	0.48	0.04	-0.77
Heckman's lambda	-20.96*	-0.68*	-5.52*
	(12.22)	(0.36)	(2.83)
Firm Fixed Effects	YES	YES	YES
Year Fixed Effects	YES	YES	YES

Standard errors are in parenthesis
**** p<0.01, ** p<0.05, * p<0.1*

The estimates show that the sample is selected on the same variables in a similar way to the main specification, except that the explanatory power for HighCashFlows is smaller but it still remains significant at the 1% level. The results for repurchases are still small and the significance that was seen in the main results table (6), disappears with when the first and tenth deciles are removed. The estimates for Heckman's lambda remain large, indicating the sample selection looks similar to the main specification, even though the level of significance has dropped to the 10% level. The estimate for the employment outcome gained significance to the 10% level, but the coefficient remains small. The results from the main specification remains robust to outliers and the effect does not change in any important way that changes the conclusions of the study.

6. Conclusions

Theory suggests that firms should redistribute cash to shareholders if they have excess cashflows without attractive investment opportunity. Previous literature made on US data have found that firms sacrifices corporate investment in favour of repurchasing own stock. The purpose of this thesis is to investigate if a similar behaviour can be found among 800 firms listed on the major stock exchanges in Sweden, Denmark, Norway and Finland between year 2000-2019. The main issue is that the sample is endogenously selected which could create a bias to the estimates. The main empirical strategy is to utilize a two-step Heckman model to account of potential sample selection bias and then estimate the value of repurchases on three different investment outcomes – capital expenditures, employment and R&D. The study finds no evidence that the included firms behave similarly to US firms by sacrificing investment in favour of repurchase on any of the investment variables. The results suggest that firms make their decisions on whether to repurchase or not, based on other variables such as available cash or lack of investment opportunity. This paper contributes to existing literature by showing that different regions are using repurchases differently and US firms may face different pressures from shareholders compared to Nordic firms. For future research I would like to see this kind of analysis expanded to additional regions and investigate if there are any heterogenous behaviours among different owner types. There's also the possibility that this study is a bit too early as we are seeing an increased trend of repurchasing but are yet to reach US level by a large margin. Given the difference in repurchasing culture, a similar effect could exist in the Nordics but hidden in cash dividends as they still are much more popular and probably of more signalling value to investors.

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Appendix.

1.

Table 8: First stage hackman results: Probit regressions for each sample identifying determinants of probability to repurchase. Test for robustness by excluding 1st and 10th decile.

	(1)	(2)	(3)
	RepurchaseDum (Capex)	RepurchaseDum (Employment)	RepurchaseDum (R&D)
High Cash Flows	0.17***	0.19***	0.28***
	(0.06)	(0.06)	(0.08)
SD Cash Flow	-0.00**	-0.00**	0.00
	(0.00)	(0.00)	(0.00)
Return on Assets	0.00	0.00*	0.00***
	(0.00)	(0.00)	(0.00)
Leverage	0.00	0.00	-0.00
	(0.00)	(0.00)	(0.00)
Size	0.07***	0.07***	0.07***
	(0.02)	(0.02)	(0.02)
Tobin's Q	0.00	0.01	0.05***
	(0.01)	(0.01)	(0.01)
Constant	-2.42***	-2.47***	-3.10***
Year Fixed Effects	YES	YES	YES

Standard errors are in parenthesis

**** p<0.01, ** p<0.05, * p<0.1*

2.

Variance inflation factor – Level of repurchases

	CAPEX	EMPLOYMENT	R&D
LEVERAGE	3.787	1.047	1.065
EBIT	3.679	1.031	1.105
CASH	1.194	1.045	1.17
SIZE	1.164	1.068	1.252
tobinsq	1.05	1.073	1.082
repurchase	1.006	1.013	1.013
Mean VIF	1.98	1.046	1.114

Variance inflation factor – Binary outcome for Repurchase

	CAPEX	EMPLOYMENT	R&D
LEVERAGE	3.601	1.033	1.05
EBIT	3.527	1.029	1.121
CASH	1.147	1.042	1.183
SIZE	1.133	1.07	1.244
tobinsq	1.044	1.066	1.089
repurdummy	1.005	1.014	1.014
Mean VIF	1.909	1.042	1.117